



KCED
Learning Augmented

24ADI003-MACHINE LEARNING

Course Objectives:

The purpose of taking this course is to:

1. Introduce the fundamental concepts of machine learning, its life cycle and ethical considerations.
2. Explore various supervised and unsupervised learning techniques and optimization strategies.
3. Examine recommendation systems, and its evaluation techniques.

Course Outcomes

- After successful completion of this course, the students shall be able to
- **CO1** Apply appropriate **data pre-processing techniques** to build machine learning models with ethical considerations. (Ap)
- **CO2** Analyze and optimize **regression models using estimation techniques, regularization, and gradient-based methods** through error analysis. (An)
- **CO3** Build and evaluate the effectiveness of **different classification models and ensemble techniques**. (E)
- **CO4** Analyze complex datasets using advanced **clustering, associative rule mining and dimensionality reduction algorithms** to uncover meaningful patterns and groupings. (An)
- **CO5** Develop **recommendation systems** to personalize user needs. (Ap)

Course Content

- **INTRODUCTION**

Introduction to Machine Learning-Types of machine learning: Supervised, Unsupervised, Semi supervised and Reinforcement Learning-Applications of machine learning in various fields- Ethics in machine learning- Fairness, accountability and interpretability -Machine learning workflow- Data Preprocessing-Feature engineering- Correlation analysis- Model training and evaluation- Model monitoring and maintenance.

- **REGRESSION MODELS**

Linear regression- Simple Regression-Least Square Estimator-Maximum Likelihood Estimator- Multiple Regression -Polynomial Regression-Performance Metrics-Bias

- **CLASSIFICATION MODELS**

Logistic Regression-Naive Bayes Classifiers-Decision Tree-K-Nearest Neighbors-Support Vector Machine – Evaluation metrics – AUC ROC- Class Imbalance –SMOTE – Cross-Validation Techniques-Ensemble Learning- Bagging- Random Forests-Boosting - AdaBoost -Gradient Boosting

- **UNSUPERVISED LEARNING**

Clustering- K-means Clustering- Gaussian Mixture Models -Hierarchical Clustering- Density-Based Clustering (DBSCAN)- Mean-Shift Clustering- Spectral Clustering- Association Rule Learning- Apriori Algorithm- FP-Growth Algorithm- Dimensionality Reduction- Principal Component Analysis (PCA)- Linear Discriminant Analysis (LDA)

- **RECOMMENDATION SYSTEMS**

Introduction to Recommendation Systems-Types- Challenges- Collaborative Filtering Techniques- User-Based- Collaborative Filtering- Item-Based- Collaborative Filtering- Matrix Factorization Techniques- Content-Based Recommendation- Hybrid Recommendation Systems- Evaluation of Recommendation Systems

Unit I – Introduction

Introduction to Machine Learning-Types of machine learning: Supervised, Unsupervised, Semi supervised and Reinforcement Learning-Applications of machine learning in various fields- Ethics in machine learning- Fairness, accountability and interpretability - Machine learning workflow- Data Preprocessing-Feature engineering- Correlation analysis- Model training and evaluation- Model monitoring and maintenance.

- **MODEL TRAINING & EVALUATION**

In supervised learning, we learn a function:

$$\mathbf{f}(\mathbf{X}) = \mathbf{y}$$

Where:

X = input features

y = output label

Goal of training:

Learn the best parameters (weights) so that model predictions match real outcomes.

• MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

We want to build an ML model to predict whether a student will be **Placed (1)** or **Not Placed (0)** using academic and skill features.

This is a **binary classification** problem.

STEP 1 — Problem Formulation (Theory)

$$f(X) = y$$

Where:

- X = input features (Marks, CodingHrs, Projects, SkillScore)
- y = output label (Placed = 1, Not Placed = 0)

Goal of training:

- Learn the best parameters (weights) so that model predictions match real outcomes.

• MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

STEP 2 — Training Dataset

Student	Marks	CodingHrs	Projects	SkillScore	Placed (y)
S1	85	5	4	88	1
S2	60	2	1	55	0
S3	75	4	3	78	1
S4	50	1	1	48	0
S5	90	6	5	92	1

Here:

- Features X = [Marks, CodingHrs, Projects, SkillScore]
- Target y = Placed

This is **labeled data**, so we use supervised learning

MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

- A **placement officer** every day, you look at student profiles like this:

Marks	CodingHrs	Projects	SkillScore
80	4	3	82

- The placement officer will ask only **one question**:
- *“Will this student get placed?”*
Yes (1) or No (0)
- This is a **binary decision**.
- So the brain is doing something like:
- “Good marks help...
Coding helps more...
Projects help a lot...
Communication matters too...”
- We are mentally **adding importance** to each factor.
- That is **exactly** what logistic regression does

MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

Logistic Regression is a mathematical model that **adds up all feature contributions** and converts the result into a **probability** between 0 and 1.

Features, Weights & Bias

(A) Features = Student Properties

- These are things we measure about a student:

Symbol	Feature	Meaning
X_1	Marks	Academic score
X_2	CodingHrs	Daily coding practice
X_3	Projects	Hands-on experience
X_4	SkillsScore	Communication + tech

- These are just **numbers**.
- But numbers alone don't tell us importance.

MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

Logistic Regression is a mathematical model that **adds up all feature contributions** and converts the result into a **probability** between 0 and 1.

Features, Weights & Bias

(B) Weights = Importance Scores

- Weights tell the model:
- “How important is this feature for placement?”

Symbol	Meaning	Intuition
W_1	Weight for Marks	How much marks matter
W_2	Weight for CodingHrs	How much coding matters
W_3	Weight for Projects	How much projects matter
W_4	Weight for SkillScore	How much skills matter

- Weights = how strongly each feature affects output.

MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

Logistic Regression is a mathematical model that **adds up all feature contributions** and converts the result into a **probability** between 0 and 1.

Features, Weights & Bias

- **(C) Bias = Base Chance**
- Bias W_0 is like:
 - “Even if the student has ZERO skills, what is the base chance of placement?”
 - It shifts everything left or right.

4. The Core Equation (The Heart of Logistic Regression)

- First, we compute a **resume strength score**:
- $$z = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4$$

MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

4. The Core Equation (The Heart of Logistic Regression)

- First, we compute a **resume strength score**:
- $z = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4$

Meaning of z

- z is:
- NOT a probability
- Just a raw score
- Can be negative or positive
- Interpretation:
- “How strong does this resume look overall?”

Symbol	Meaning
(w_0)	Bias (baseline placement tendency)
(w_1, w_2, w_3)	Weights (importance of features)
(x_1, x_2, x_3)	Marks, CodingHrs, Projects

5. Why We Cannot Use z Directly

Problem:

$z = 15$ (not a probability)

$z = -8$ (not a probability)

But we need output like:

$0.73 \rightarrow 73\%$ chance

$0.12 \rightarrow 12\%$ chance

So we need a **converter**.

MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

6. Enter the Sigmoid Function

- Sigmoid squashes any number into (0, 1):

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

What Sigmoid Does

z value	$\sigma(z)$	Meaning
-10	0.00004	Almost no chance
-2	0.12	Low chance
0	0.5	50-50
+2	0.88	High chance
+10	0.999	Almost sure

sigmoid function is like:
A strict HR filter
Bad resumes → thrown out
Good resumes → selected
Borderline → confused

MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

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MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

7. Final Logistic Regression Model

- Putting it all together:

$$P(\text{Placed}) = \frac{1}{1 + e^{-(w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4)}}$$

8. What Training Actually Does

- Students always ask:
- “Sir, where do these weights come from?”

Training = Learning Weights

The model:

- Starts with random weights
- Makes predictions
- Calculates error
- Adjusts weights
- Repeats thousands of times

Goal:

- Find weights that make predictions close to reality.

MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

STEP 3 — Choice of Model: Logistic Regression

Why logistic regression?

- Output is binary
- Produces probability values
- Interpretable weights

Model Equation

First compute a linear score z :

$$Z = W_0 + W_1X_1 + W_2X_2 + W_3X_3 + W_4X_4$$

Then convert z into probability using sigmoid:

$$P(y = 1 | X) = \sigma(z) = 1 / (1 + e^{(-z)})$$

Why sigmoid?

- Maps any real number to range (0, 1)
- Interpretable as probability

Decision rule:

- If $P \geq 0.5 \rightarrow$ predict Placed
- Else $\rightarrow$ predict Not Placed

MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

• STEP 4 — Model Training

Training means:

- Start with random weights
- Compute predictions
- Measure error using loss function
- Update weights using gradient descent

Loss Function (Binary Cross-Entropy)

$$L = -[y \log(p) + (1 - y) \log(1 - p)]$$

Where:

- y = true label
- p = predicted probability
- Gradient descent updates weights to minimize L .

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MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

STEP 5 — Learned Model (After Training)

Assume training produced:

$$z = -7 + 0.06(\text{Marks}) + 0.8(\text{CodingHrs}) + 0.7(\text{Projects}) + 0.05(\text{SkillScore})$$

Interpretation of Weights

Feature	Weight	Description
Marks	0.06	Small positive effect
CodingHrs	0.8	Strong positive effect
Projects	0.7	Strong positive effect
SkillScore	0.05	Small effect

Interpretation:

Coding practice and projects dominate placement decisions.

MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

STEP 6 — Prediction (New/future value)

Given Student

Feature	Value
Marks	80
CodingHrs	4
Projects	3
SkillScore	82

• **Step 6A — Compute Linear Score z**

- $z = -7 + 0.06(80) + 0.8(4) + 0.7(3) + 0.05(82)$
- $= -7 + 4.8 + 3.2 + 2.1 + 4.1$
- $= 7.2$

Interpretation:

- **High resume strength score.**

MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

STEP 6 — Prediction (New/future value)

Given Student

Feature	Value
Marks	80
CodingHrs	4
Projects	3
SkillScore	82

Step 6B — Apply Sigmoid Function

$$P(\text{Placed}) = 1 / (1 + e^{(-7.2)}) \approx 0.999$$

Interpretation:

Student has 99.9% probability of placement.

MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

STEP 7 — Borderline Case

Feature	Value
Marks	55
CodingHrs	1
Projects	1
SkillScore	50

$$z = -7 + 3.3 + 0.8 + 0.7 + 2.5$$

$$z = 0.3$$

$$P(\text{Placed}) = 1 / (1 + e^{(-0.3)}) \approx 0.574$$

Interpretation:

Borderline candidate — improvement in coding/projects can flip result.

MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

STEP 8 — Model Evaluation

- After training, we must test model on unseen data.
- We use a **confusion matrix**.

STEP 9 — Confusion Matrix (Test Data)

Actual \ Pred	No	Yes
No	18	2
Yes	3	17

Where:

- TP = 17
- TN = 18
- FP = 2
- FN = 3

MODEL TRAINING & EVALUATION

Scenario: Campus Placement Prediction

Actual \ Pred	No	Yes
No	18	2
Yes	3	17

Where:

- TP = 17
- TN = 18
- FP = 2
- FN = 3

STEP 10 — Evaluation Metrics

Accuracy

$$\text{Accuracy} = (\text{TP} + \text{TN}) / \text{Total}$$

$$= (17 + 18) / 40 = 35 / 40 = 87.5\%$$

Meaning:

Model predicts correctly 87.5% of the time.

Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$= 17 / (17 + 2) = 17 / 19 = 0.89$$

Meaning:

When model predicts "Placed", it is correct 89% of the time.

Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$= 17 / (17 + 3) = 17 / 20 = 0.85$$

Meaning:

Model identifies 85% of actually placed students.

F1-Score

$$\text{F1} = 2\text{PR} / (\text{P} + \text{R})$$

$$= 2(0.89 \times 0.85) / (0.89 + 0.85)$$

$$\approx 0.87$$

Meaning:

Balanced measure of precision and recall.

MODEL TRAINING & EVALUATION

Model training learns weights that map features to placement probability.

- Logistic regression:
- Computes z
- Applies sigmoid
- Produces probability

Model evaluation:

- Uses confusion matrix
- Computes accuracy, precision, recall, F1

Together they ensure:

- **The model is both intelligent and trustworthy.**

MODEL MONITORING & MAINTENANCE

Why Model Monitoring & Maintenance Are Needed

- In Machine Learning, we build models using **historical data**. Implicitly, we assume:
- The future will behave like the past.
- Mathematically, we assume:

$$P_{train}(X, y) \approx P_{future}(X, y)$$

- But in real life, this assumption is fragile.

Why the Assumption Breaks

- Because the world is **non-stationary**:
- • Students learn new skills
- • Companies change hiring criteria
- • New technologies appear
- • Old skills become obsolete
- • Online learning platforms rise
- So the **data-generating process itself changes**.

MODEL MONITORING & MAINTENANCE

Fundamental Principle

- A model trained on past data remains valid **only while** the statistical properties and relationships in data remain unchanged.

Once this breaks, predictions become:

- Biased
- Overconfident
- Inaccurate
- Misleading for decision-making

A GPS map from 2015 is unreliable in 2026 because:

- • New roads were built
- • Old roads were closed
- • Traffic rules changed

Similarly:

- A model trained in 2023 becomes unreliable in 2026.

MODEL MONITORING & MAINTENANCE

Model Monitoring

- **Model Monitoring** is the continuous and systematic process of:
 - Tracking model performance metrics
 - Observing input data distributions
 - Detecting abnormal behaviour
 - Identifying performance degradation in a deployed machine learning system.

MODEL MONITORING & MAINTENANCE

Model Monitoring

What is Monitored?

1. Output Quality

- Accuracy
- Precision
- Recall
- F1-score

Purpose:

- Are predictions still correct?

2. Input Data Distribution

- Mean, variance, percentiles
 - Category frequencies
 - Feature correlations

Purpose:

Is today's data similar to training data?

3. Prediction Confidence

- Probability scores
- Entropy of predictions

Purpose:

Is the model becoming uncertain?

4. Error Rates

- False positives
- False negatives

Purpose:

Are costly mistakes increasing?

5. Business KPIs

- Placement success rate
- Employer satisfaction
- Student complaints

Purpose:

Is the model still useful in practice?

MODEL MONITORING & MAINTENANCE

Model Monitoring

A deployed ML model is a **living system** that must be observed continuously.

Deployment Scenario (Baseline Model)

- A campus placement prediction model is trained in **2023**.

Training Features Used

Feature	Meaning
Marks	Academic performance
C	Programming foundation
Java	Object-oriented skills
Aptitude	Logical ability

• **High accuracy** →
Good overall correctness

• **High precision** →
Few false placements

Training Performance

Metric	Value
Accuracy	88%
Precision	0.90
Recall	0.86
F1-Score	0.88

• **High recall** → **Most good students detected**
So:
The model is both accurate and balanced.

MODEL MONITORING & MAINTENANCE

Model Monitoring

Model Drift?

- After deployment, **reality starts evolving**.
- Two distinct types of drift appear.

A) Data Drift

- **Data Drift** occurs when the statistical distribution of input features changes, even if the meaning of the target remains the same.
- Formally:

$$P_{train}(X) \neq P_{live}(X)$$

Example

Feature	2023 Mean	2026 Mean
Marks	72	68
CodingHrs	1.2	3.8
GitHubStars	0	25

- Students now code more
- GitHub has become important
- The model never learned GitHubStars

So:

The model is being asked to make decisions using **unfamiliar data**.

MODEL MONITORING & MAINTENANCE

Model Monitoring

Model Drift?

- After deployment, **reality starts evolving**.
- Two distinct types of drift appear.

B) Concept Drift

- **Concept Drift** occurs when the *meaning of patterns* changes.
- Formally:

$$P_{train}(y | X) \neq P_{live}(y | X)$$

Example

In 2023:

- High Java + Aptitude → Placement likely

In 2026:

- High ML + GitHub + Kaggle → Placement likely
- Even if Marks and Java remain high:
- They no longer guarantee placement.
- So:
- The rule learned by the model has become outdated.

MODEL MONITORING & MAINTENANCE

Model Monitoring

5. Live Monitoring Results (Observed Failure)

- Monitoring 300 students in 2026.

Confusion Matrix

Actual \ Predicted	No	Yes
No	120	40
Yes	78	62

- Accuracy**

- Accuracy = $\frac{TP+TN}{Total} = \frac{62+120}{300} = 60.7\%$

- Many false negatives → Good students rejected
- Many false positives → Weak students selected

So:

- The model is now harmful.

MODEL MONITORING & MAINTENANCE

Model Maintenance

Model Maintenance is the systematic process of:

- Updating training data
- Adding or removing features
- Retraining model parameters
- Re-evaluating performance

to restore or improve model accuracy and relevance.

Maintenance Goals

- Restore predictive accuracy
- Capture new market trends
- Reduce wrong decisions
- Maintain trust in the system

MODEL MONITORING & MAINTENANCE

7. Step-wise Maintenance Actions

Action 1: Add New Features

Feature	Why Needed
PythonScore	Industry demand
MLScore	AI jobs growth
GitHubStars	Real coding signal
KaggleScore	Practical ML skill

Action 2: Collect New Training Data

Marks	CodingHrs	Python	ML	GitHubStars	KaggleScore	Placed
75	3	8	7	45	350	1
62	4	6	5	12	120	1
58	1	3	2	0	0	0

MODEL MONITORING & MAINTENANCE

7. Step-wise Maintenance Actions

Action 3: Retrain Model

The model relearns:

- New weights
- New decision boundaries
- New feature importance

8. Retrained Model Evaluation

Confusion Matrix

Actual \ Predicted	No	Yes
No	130	20
Yes	18	132

- **Accuracy**
- $\frac{130+132}{300} = 87.3\%$

Interpretation

- The model has adapted to new reality.

MODEL MONITORING & MAINTENANCE

9. Monitoring Threshold Example

Accuracy Trend

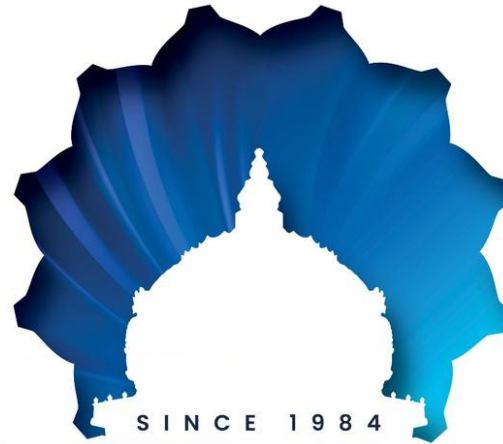
Month	Accuracy
Jan	88%
Feb	86%
Mar	83%
Apr	79%
May	74%
Jun	68%

- **Rule**
- Retrain *if Accuracy* < 80%
- So retraining starts in **April**.

MODEL MONITORING & MAINTENANCE

Term	Description
Model Monitoring	Tracking performance & data
Data Drift	Change in X
Concept Drift	Change in $X \rightarrow y$
Model Maintenance	Updating model after drift

- Model Training builds intelligence.
- Model Monitoring protects intelligence.
- Model Maintenance refreshes intelligence.



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